

Joins

Agenda and reminders

- HW 2 due Friday on **GitHub classroom**
 - Commit early and often, let me know if you have any technical problems
 - Make sure to submit both the `.qmd` **and** `.md` files
- Today: combining data from different tables

Data stored in multiple tables

The `nycflights13` package contains information on flights from NYC airports in 2013. The data is stored across several data frames:

- `airlines`: information on each airline
- `airports`: information on each airport
- `flights`: information on each flight
- `planes`: information on each plane
- `weather`: hourly weather data

Question: What is the advantage of storing this data in multiple tables, instead of one BIG table?

Data stored in multiple tables

- Databases often contain different tables to store different information
- For example, a healthcare database could contain the following tables:
 - patients
 - doctors
 - offices
 - insurance

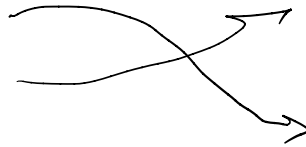
Joining tables

1 patients

	age	insurance	provider
1	23	Aetna	Dr. Zhang
2	47	BCBS	Dr. Foyle
3	38	Medicaid	Dr. Zhang

1 doctors

	provider	location
1	Dr. Foyle	Greensboro
2	Dr. Renard	Winston-Salem
3	Dr. Zhang	Winston-Salem



I want to add location information to the patient table.
What should the resulting table look like?

age	insurance	provider	location
23	Aetna	Dr. Zhang	W-S
47	BCBS	Dr. Foyle	Greensboro

Left join

```
1 patients
```

	age	insurance	provider
1	23	Aetna	Dr. Zhang
2	47	BCBS	Dr. Foyle
3	38	Medicaid	Dr. Zhang

```
1 doctors
```

	provider	location
1	Dr. Foyle	Greensboro
2	Dr. Renard	Winston-Salem
3	Dr. Zhang	Winston-Salem

```
1 patients |>
```

```
2   left_join(doctors, join_by(provider))
```

has to link the
tables together

	age	insurance	provider	location
1	23	Aetna	Dr. Zhang	Winston-Salem
2	47	BCBS	Dr. Foyle	Greensboro
3	38	Medicaid	Dr. Zhang	Winston-Salem

Left join

```
1 patients |>  
2   left_join(doctors, join_by(provider))
```

	age	insurance	provider	location
1	23	Aetna	Dr. Zhang	Winston-Salem
2	47	BCBS	Dr. Foyle	Greensboro
3	38	Medicaid	Dr. Zhang	Winston-Salem

- Left joins are useful for adding additional information to a table
- Left joins (generally) keep the same rows as the initial dataframe (`patients`), and add more columns
- `join_by` specifies how to link the tables

Joining tables

flights |>
left_join(weather,
join_by(origin,
time_hour))

Flights information:

A tibble: 3 × 5

	time_hour <dtm>	origin <chr>	dest <chr>	tailnum <chr>	carrier <chr>
1	2013-01-01 05:00:00	EWR	IAH	N14228	UA
2	2013-01-01 05:00:00	LGA	IAH	N24211	UA
3	2013-01-01 05:00:00	JFK	MIA	N619AA	AA

Weather information

A tibble: 3 × 4

	origin <chr>	time_hour <dtm>	temp <dbl>	wind_speed <dbl>
1	EWR	2013-01-01 01:00:00	39.0	10.4
2	EWR	2013-01-01 02:00:00	39.0	8.06
3	EWR	2013-01-01 03:00:00	39.0	11.5

Question: What if I want to get information about the weather for each flight?

Left joins

```
1 flights |>
2   left_join(weather, join_by(origin, time_hour))
```

A tibble: 6 × 7

	time_hour <dtm>	origin <chr>	dest <chr>	tailnum <chr>	carrier <chr>	temp <dbl>	wind_speed <dbl>
1	2013-01-01 05:00:00	EWR	IAH	N14228	UA	39.0	12.7
2	2013-01-01 05:00:00	LGA	IAH	N24211	UA	39.9	15.0
3	2013-01-01 05:00:00	JFK	MIA	N619AA	AA	39.0	15.0
4	2013-01-01 05:00:00	JFK	BQN	N804JB	B6	39.0	15.0
5	2013-01-01 06:00:00	LGA	ATL	N668DN	DL	39.9	16.1
6	2013-01-01 05:00:00	EWR	ORD	N39463	UA	39.0	12.7

Joining with different names

Suppose our tables looked like this:

1 patients				1 doctors	
	age	insurance	provider	name	location
1	23	Aetna	Dr. Zhang	1 Dr. Foyle	Greensboro
2	47	BCBS	Dr. Foyle	2 Dr. Renard	Winston-Salem
3	38	Medicaid	Dr. Zhang	3 Dr. Zhang	Winston-Salem

How would we specify the columns to link the tables?

link
join-by (provider == name)

Joining with different names

Suppose our tables looked like this:

```
1 patients
```

	age	insurance	provider
1	23	Aetna	Dr. Zhang
2	47	BCBS	Dr. Foyle
3	38	Medicaid	Dr. Zhang

```
1 doctors
```

	name	location
1	Dr. Foyle	Greensboro
2	Dr. Renard	Winston-Salem
3	Dr. Zhang	Winston-Salem

```
1 patients |>  
2 left_join(doctors, join_by(provider == name))
```

	age	insurance	provider	location
1	23	Aetna	Dr. Zhang	Winston-Salem
2	47	BCBS	Dr. Foyle	Greensboro
3	38	Medicaid	Dr. Zhang	Winston-Salem

provider (from patients table)
match name (from doctors)

Another join

Patients in the system:

1	patients
---	----------

	age	insurance	provider
1	23	Aetna	Dr. Zhang
2	47	BCBS	Dr. Foyle
3	38	Medicaid	Dr. Zhang
4	54	Humana	Dr. Renard

Accepted insurance:

1	insurance
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	company	phone
1	Anthem	800-676-2583
2	BCBS	877-258-3334
3	Kaiser	800-810-4766
4	Medicaid	877-201-3750

Suppose I want insurance information only for the patients who have an accepted insurance. What should the final table look like?

age	insurance	provider	phone
47	BCBS	...	...
38	Medicaid	...	...

Inner join

Patients in the system:

```
1 patients
```

	age	insurance	provider
1	23	Aetna	Dr. Zhang
2	47	BCBS	Dr. Foyle
3	38	Medicaid	Dr. Zhang
4	54	Humana	Dr. Renard

Accepted insurance:

```
1 insurance
```

	company	phone
1	Anthem	800-676-2583
2	BCBS	877-258-3334
3	Kaiser	800-810-4766
4	Medicaid	877-201-3750

```
1 patients |>
```

```
2   inner_join(insurance, join_by(insurance == company))
```

	age	insurance	provider	phone
1	47	BCBS	Dr. Foyle	877-258-3334
2	38	Medicaid	Dr. Zhang	877-201-3750

link insurance & company
columns

keep only rows for which insurance / company
appears in both tables

Class activity

- Work with a neighbor on the class activity
- At the end of class, submit your work as an HTML file on Canvas (one per group, list all your names)

For next time, read:

- Chapter 26.2 in *R for Data Science*